

ESP32

v1.0

MIT

License

ESP32 HARNESS

Advanced Pentesting, Audit & Telemetry Firmware

2.4 GHz Research | RF24 Experimentation | BLE Scanning | Forensic Capture

2.4 GHz

Wi-Fi Engine

RF24

NRF24L01+

BLE

Bluetooth

PCAP

Forensic

MQTT

Telemetry

GPIO

Hardware

ESP32-WROOM-32

Dual-core 240-MHz
520 KB SRAM

NRF24L01+ PA/LNA

2.4 GHz ISM
+20 dBm

MicroSD Module

FAT32/exFAT
Up to 32 GB

SSD1306 OLED

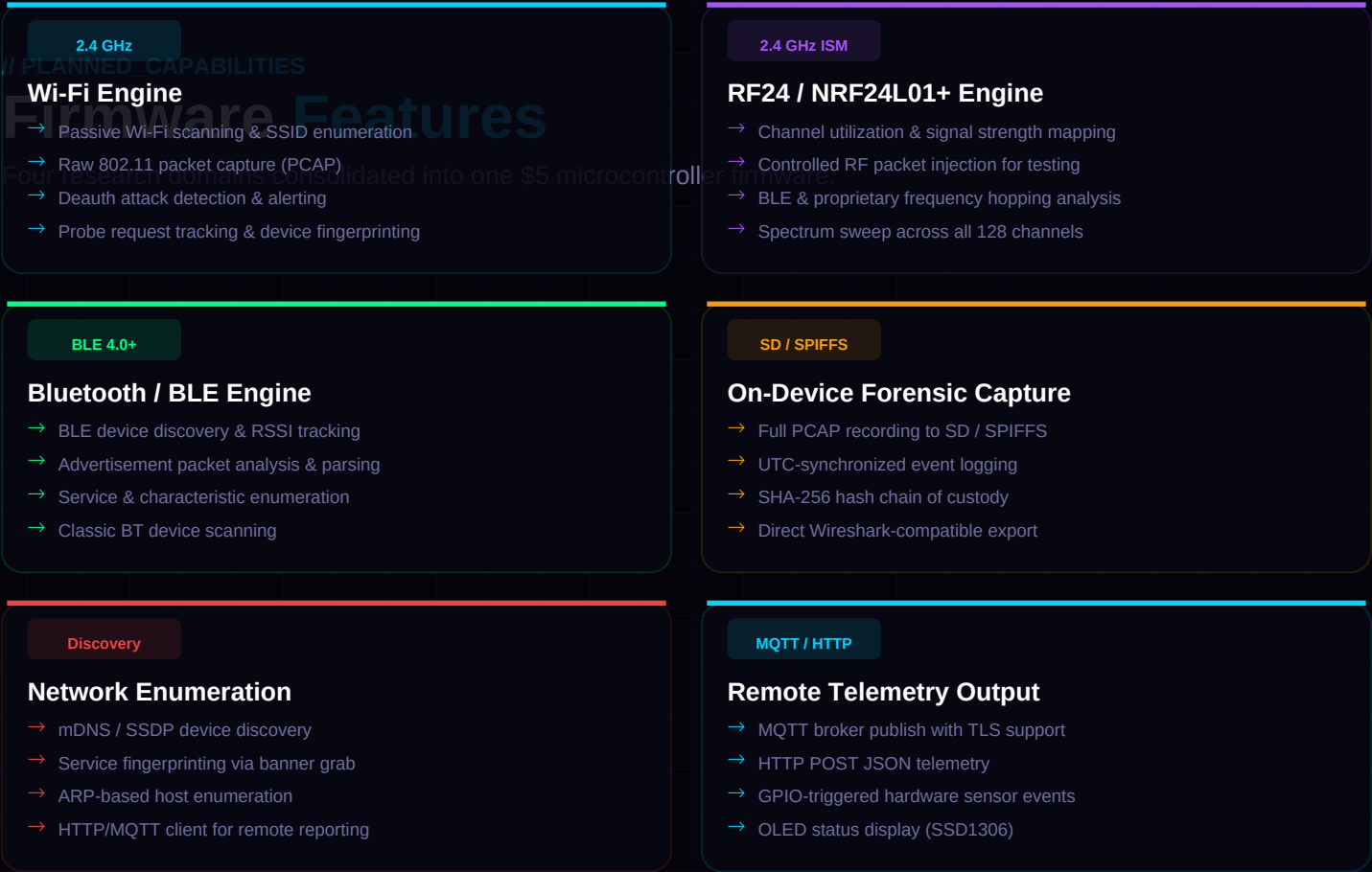
128x64 Mono
I2C 400kHz

LiPo 3.7V

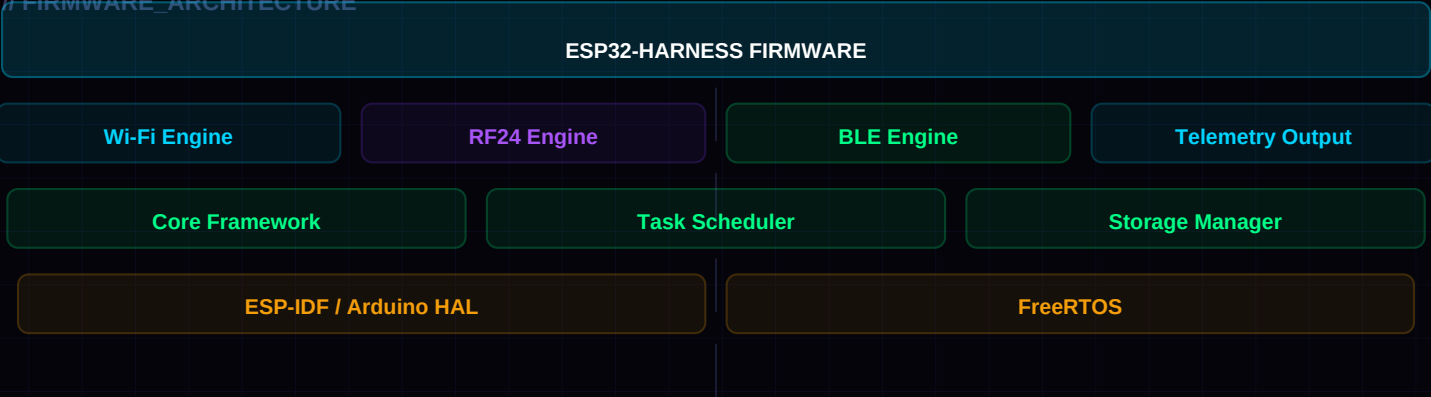
1000mAh
~4 hrs runtime

// CAPABILITIES

- **Wi-Fi Research**
802.11 frame capture, deauth monitoring, probe tracking
- **Packet Injection**
Controlled RF transmission for protocol testing
- **Forensic Capture**
Full PCAP to SD, UTC timestamps, Wireshark export
- **RF24 Spectrum**
ISM band analysis, channel mapping, signal visualization
- **BLE Enumeration**
Bluetooth Low Energy device discovery & RSSI tracking
- **GPIO Events**
Hardware sensors, MQTT/HTTP telemetry output



// FIRMWARE_ARCHITECTURE



// QUICK_START

Get Started

From zero to flashing in under 5 minutes.

01 Clone

```
git clone https://github.com/ykrishhh/ESP32-HARNESS.git
```

02 Install

```
pip install platformio && pio pkg install
```

03 Configure

```
Edit config.h — set RF24_ENABLED, SD_ENABLED, MQTT_TELEMETRY
```

04 Flash

```
pio run --target upload
```

05 Monitor

```
pio device monitor --baud 115200
```

// BILL_OF_MATERIALS

Minimum viable setup costs under \$10. Full research rig under \$25.

Component	Model / Spec	Purpose	Cost	Status
ESP32 Module	ESP32-WROOM-32 / DevKitC	Main controller	~\$3-5	Required
NRF24L01+ PA/LNA	2.4 GHz transceiver + antenna	RF24 experimentation	~\$2-4	Optional
MicroSD Module	SPI interface, FAT32	PCAP capture storage	~\$1-2	Optional
OLED Display	SSD1306 128x64, I2C	Real-time status display	~\$2-3	Optional
LiPo Battery	3.7V 1000mAh + TP4056	Portable field operation	~\$3-5	Optional

// COMPONENT_MATRIX

Component	Interface	Protocol	Speed
ESP32-WROOM-32	UART / USB	Serial	115200 baud
NRF24L01+ PA/LNA	SPI	SPI	10 MHz
MicroSD Module	SPI	SPI	25 MHz
OLED SSD1306	I2C	I2C	400 kHz
LiPo Battery	ADC	3.7V	nominal